

# Md. Farhan Mahtab

Dhaka, Bangladesh

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## Education

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**Bangladesh University of Engineering and Technology**

April, 2019 – June, 2024

*B.Sc in Computer Science and Engineering | CGPA: 3.83/4.00*

*Dhaka, Bangladesh*

## Research Interests

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Trustworthy AI & LLM Security · Formal Methods · IoT & Systems Security · Privacy-Preserving Machine Learning · Federated Learning · Intelligent Systems

## Research Experience

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**Systematic Vulnerability Analysis of Matter SDK**

August, 2026 – Present

*University of Texas Dallas*

*Dallas, Texas, USA*

- Investigating security vulnerabilities in the Matter SDK through formal models generated from system specifications using large language models (LLMs).
- Exploring an LLM-assisted formal modeling approach for systematically analyzing the security properties of software specifications.
- Working at the intersection of LLMs, formal methods, IoT security, and automated vulnerability analysis

**FedCrime: A Federated Learning Based Crime Prediction Model**

July, 2023 – May, 2024

*Bangladesh University of Engineering and Technology*

*Dhaka, Bangladesh*

- Undergraduate Thesis Supervised by Dr. Tanzima Hashem
- Developed a privacy-preserving crime prediction approach using federated learning over private crime data
- Investigated how distributed learning can improve predictive modeling while avoiding direct sharing of sensitive data
- Explored the application of federated learning to privacy-sensitive predictive analytics

**Myers-Briggs Personality Type Identification Using LLM**

October, 2023 – February, 2024

*Bangladesh University of Engineering and Technology*

*Dhaka, Bangladesh*

- Fine-tuned a large language model to generate embeddings from social media text
- Developed a classification model using LLM-derived text embeddings to identify the INTJ personality type
- Investigated the use of LLM representations for downstream classification of personality-related textual information

## Job Experience

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**Software Engineer**

June, 2024 – Present

*Samsung R&D Institute Bangladesh*

*Dhaka, Bangladesh*

- Developed and maintained software for the Tizen Developer Interface and Tizen developer ecosystem
- Optimized the Tizen emulator based on the open-source QEMU project, focusing on emulator performance

- Developed Rust-based GUI tools for the Tizen SDK
- Developed an AI agent for the Tizen SDK, combining software development tooling with AI-based capabilities
- Gained hands-on experience in systems software, emulation, developer tooling, operating-system ecosystems, and AI-assisted software development

#### **Intern**

May, 2023 - June, 2023

*ERA-InfoTech Limited*

*Dhaka, Bangladesh*

- Developed an unsupervised machine learning model for classification of bank transactions
- Applied machine learning techniques to financial transaction data for automated categorization

### *Research & Technical Areas*

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#### **Trustworthy AI**

- LLM-assisted Security Analysis · LLMs · Formal Modeling · Vulnerability Analysis

#### **Machine Learning**

- Federated Learning · Privacy-Preserving ML · Representation Learning · Text Classification · Unsupervised Learning

#### **Systems & Security**

- IoT Security · Matter SDK · Tizen · QEMU · Emulator Optimization · Systems Software

#### **Development**

- Rust · Software Engineering · SDK & Developer Tooling · AI Agents

### *Awards & Honors*

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#### **Icon Engineer**

*Samsung R&D Institute Bangladesh*

2026

#### **RISE Student Research Grant**

*Bangladesh University of Engineering and Technology*

2023

#### **Dean's Award**

*Bangladesh University of Engineering and Technology*

2021